Supporting Text S1 — derivations behind Section 3.3

draft, 2026-10-20, untracked working file

Notes (delete before pasting): checked against Ribe (2001) on 2026-10-20; equation numbers cited for Ribe are his. S1.1 is the shear endmember, the derivation the submitted paper's Text S1 gave for the H^1 scaling, written here as an endmember rather than as the result. S1.2 is the moment-balance route that Section 3.3 now rests on. Coordinates and signs match the extraction and Tests A and C (s down-dip, n across the slab, $K = d\theta/ds$). One difference from the submitted Text S1 is deliberate: the old S1 wrote the shear stress as $\eta\, dv_n/ds$ and evaluated dv_n/ds as the derivative of the component $\mathbf{v}\cdot\hat{\mathbf{n}}$, which equals $\omega + Kv_s$ (Equation S2 below); the Kv_s part is the rotation of the coordinate basis, not a strain, so the shear-mode stress is $\eta\omega$. This changes the coefficients in Equation S6 relative to the old Equation 8 but not the dependence on η, H, K and v_s.

Text S1. Derivation of the in-slab shear force gradient

S1.0 Coordinates

We write the slab in coordinates aligned with its midsurface: s is arc length along the midsurface, positive down-dip; n is the distance from the midsurface; $\theta(s)$ is the dip; and $K = d\theta/ds$ is the curvature. With x to the right and y upward, the unit vectors along and across the slab are $\hat{\mathbf{t}} = (\cos\theta, -\sin\theta)$ and $\hat{\mathbf{n}} = (\sin\theta, \cos\theta)$, so that $v_s = v_x\cos\theta - v_y\sin\theta$ and $v_n = v_x\sin\theta + v_y\cos\theta$. Because θ varies along the slab, the unit vectors rotate with s:

$$\frac{d\hat{\mathbf{t}}}{ds} = -K\hat{\mathbf{n}}, \qquad \frac{d\hat{\mathbf{n}}}{ds} = K\hat{\mathbf{t}}. \qquad (S1)$$

S1.1 The through-thickness shear endmember ($dQ_s/ds \propto H$)

The rotation rate of the slab midsurface is the rate at which its tangent turns, $\omega = \hat{\mathbf{n}}\cdot\partial\mathbf{v}/\partial s$. Written in terms of the velocity components,

$$\omega = \frac{\partial(\mathbf{v}\cdot\hat{\mathbf{n}})}{\partial s} - \mathbf{v}\cdot\frac{d\hat{\mathbf{n}}}{ds} = \frac{\partial v_n}{\partial s} - Kv_s, \qquad (S2)$$

which is the quantity that appears in Equation 6 of the main text. The second term is there because the component v_n is measured against a direction that itself rotates along the slab; it is not a deformation. In the same way, the rotation rate of a material line drawn across the slab is $-\partial v_s/\partial n$. The shear strain rate at the midsurface is the difference between the rotation of a cross-section and the rotation of the midsurface it was normal to,

$$2\dot{\varepsilon}_{sn} = \frac{\partial v_s}{\partial n} + \omega, \qquad (S3)$$

and the shear stress is $\tau_{sn} = 2\eta\,\dot{\varepsilon}_{sn}$. Equation S3, with $\tau_{sn} = 2\eta\,\dot{\varepsilon}_{sn}$, is the exact midsurface shear stress of a curved viscous sheet (Ribe, 2001, the expression for σ_{zs} below his equation 2.1, with his $\omega = w_0' + Ku_0$; his K is positive with the centre of curvature on the $+z$ side, ours on the $-n$ side, which is the only difference). A rigid rotation of a curved slab, $v_s = \Omega(R + n)$ with $R = 1/K$, gives $\partial v_s/\partial n = \Omega$ and $\omega = -\Omega$, so Equation S3 vanishes as it must.

The shear endmember is the assumption that cross-sections translate without rotating, $\partial v_s/\partial n = 0$. All of the midsurface rotation must then be accommodated by shearing the section:

$$\tau_{sn} = \eta\omega, \qquad Q_s = \int \tau_{sn}\, dn = \eta H\omega. \qquad (S4)$$

For a slab that moves down-dip through a geometry that is steady in the slab frame, the velocity is along the slab, $\mathbf{v} = v_s\hat{\mathbf{t}}$, and Equation S2 with Equation S1 gives

$$\omega = v_s\,\hat{\mathbf{n}} \cdot \frac{d\hat{\mathbf{t}}}{ds} = -Kv_s. \qquad (S5)$$

The shear force is then $Q_s = -\eta HKv_s$, and its gradient along the slab is

$$\frac{dQ_s}{ds} = -\eta H\left(K\frac{dv_s}{ds} + v_s\frac{dK}{ds}\right). \qquad (S6)$$

If K and v_s vary over a common length L along the slab, $|dQ_s/ds| \sim 2\eta HKv_s/L$, i.e.

$$\frac{dQ_s}{ds} \propto \eta HKv_c \qquad (S7)$$

with $v_s \approx v_c$ and a prefactor absorbing H/L. This is the scaling of the submitted manuscript. It is linear in plate thickness because the shear stress is assumed uniform across the slab, so that doubling H doubles the shear force at the same stress.

The models do not deform this way, and thin-sheet theory says they should not: for a sheet much thinner than the length over which it deforms, the along-slab velocity varies across the sheet as $-\omega n$ at leading order (Ribe, 2001, equation 4.6a), which is the Kirchhoff behaviour. The measured $\partial v_s/\partial n$ equals $-\omega$ to within a few per cent across the whole suite (Figure S-bend b), so the two terms of Equation S3 cancel, the constitutive shear stress is several times smaller than $\eta|\omega|$, and the shear force gradient scales as H^3 rather than H across the $H = 70$, 100 and 130 km models (Figure S-varH). Section 3.3 therefore uses the bending endmember, derived next.

S1.2 The bending endmember: moment balance and the thin-sheet moment

When cross-sections rotate with the midsurface ($\partial v_s/\partial n = -\omega$), Equation S3 gives $\dot{\varepsilon}_{sn} \approx 0$ at leading order, and the shear stress can no longer be obtained from the constitutive law. It is fixed instead by equilibrium.

Moment balance. The along-slab component of the momentum equation is

$$\frac{\partial \sigma_{ss}}{\partial s} + \frac{\partial \sigma_{sn}}{\partial n} + \rho g_s = 0 \qquad (S8)$$

up to terms in $K\sigma_{sn}$ that come from the curvature of the coordinates (Ribe, 2001, equation 2.1b); σ_{ss} is the total along-slab normal stress (deviatoric stress minus pressure), $\sigma_{sn} = \tau_{sn}$, and g_s is the down-dip component of gravity. Multiplying by n and integrating across the slab from $n = -H/2$ to $+H/2$ (the curvature terms cancel exactly in this weighted integral):

$$\int n \, \frac{\partial \sigma_{ss}}{\partial s} \, dn = \frac{dM}{ds}, \qquad M = \int n \, \sigma_{ss} \, dn,$$

the bending moment;

$$\int n \, \frac{\partial \sigma_{sn}}{\partial n} \, dn = \Big[n \, \tau_{sn} \Big]_{-H/2}^{+H/2} - \int \tau_{sn} \, dn = C - Q_s, \qquad C = \frac{H}{2} \left[\tau_{sn}\Big(+\tfrac{H}{2}\Big) + \tau_{sn}\Big(-\tfrac{H}{2}\Big) \right],$$

the couple exerted by the mantle shear tractions on the two faces; and

$$\int n \, \rho g_s \, dn = 0$$

when the density is uniform across the slab (the lithostatic part of the pressure balances the reference density, and the excess density is symmetric about the midsurface). Hence

$$Q_s = \frac{dM}{ds} + C. \qquad (S9)$$

Equation S9 is the exact torque balance of thin viscous sheet theory (Ribe, 2001, equation 2.4; Buffett, 2006, equation 5, where the face tractions are neglected). Retaining the curvature of the slab in the volume element, $(1 + Kn)\, dn$, adds a couple from the slab's own weight, $\frac{1}{12} K H^3 \Delta\rho \, g_s$ (Ribe, 2001, equation 2.4; it appears with H in Buffett's equation 5, a misprint, since his equation 7 and the dimensions require H^3); in the models it is 2 % of Q_s at the median and 13 % at the 90th percentile, and it is neglected here. The measured partition of Q_s in the models is $dM/ds \approx 0.86\, Q_s$ and $C \approx -0.12\, Q_s$ at the median (Figure S-bend c), so Q_s is carried almost entirely by the moment gradient.

Constitutive relation for the moment. With cross-sections rotating with the midsurface, the along-slab velocity varies linearly across the slab, $v_s(s, n) = v_s(s, 0) - n\, \omega(s)$, and so does the along-slab strain rate:

$$\dot{\varepsilon}_{ss} = \Delta + n\dot{\kappa}, \qquad \dot{\kappa} = -\frac{\partial \omega}{\partial s} = \frac{\partial^2 v_s}{\partial s \, \partial n}, \qquad (S10)$$

where Δ is the stretching rate of the midsurface and $\dot{\kappa}$ is the bending rate, $\Omega = -\omega'$ in Ribe (2001). The rate of change of curvature of the material midsurface is $\Omega + K\Delta$ (Ribe, 2001, after equation 3.1); the $K\Delta$ part, curvature times stretching, is small for slabs that do not stretch, and the moment it induces is dropped here, as in Ribe's own thin-sheet equations (his 4.10b versus 4.11). The slab is incompressible and in plane strain, so $\dot{\varepsilon}_{nn} = -\dot{\varepsilon}_{ss}$, and across a thin sheet the normal

stress $\sigma_{nn} = -p + 2\eta\,\dot{\varepsilon}_{nn}$ is small compared with the bending stresses, which fixes the pressure at $p = -2\eta\,\dot{\varepsilon}_{ss}$ (the linear-in-n pressure $p = -2\eta\Omega n$ of Ribe, 2001, equation 4.6b). The along-slab normal stress is therefore

$$\sigma_{ss} = -p + 2\eta\,\dot{\varepsilon}_{ss} = 4\eta\,\dot{\varepsilon}_{ss} \qquad (S11)$$

(Buffett, 2006, equation 12). Substituting Equations S10 and S11 into the definition of M, the Δ term integrates to zero and

$$M = 4\eta\dot{\kappa}\int n^2\,dn = \frac{\eta H^3\dot{\kappa}}{3}, \qquad (S12)$$

which is equation 4.8b of Ribe (2001), equation 16 of Ribe (2010) and equation 13 of Buffett (2006), the latter with $\dot{\kappa} = u_0\,dK/ds$ for a steady shape translating at speed u_0. This is Equation 10 of the main text. In the models the measured M is within 10 % of Equation S12 at the median (Figure S-bend d).

The shear force gradient. Differentiating Equation S9 and neglecting C and along-slab variations in η and H,

$$\frac{dQ_s}{ds} \approx \frac{d^2M}{ds^2} = \frac{\eta H^3}{3}\,\frac{d^2\dot{\kappa}}{ds^2}. \qquad (S13)$$

For a slab moving through a curved geometry, $\omega \approx -Kv_s$ (Equation S5; in Ribe, 2001, equation 6.7, ω is the rate of change of the dip following the material, which for a slab sliding along a fixed shape is $v_s\,d\theta/ds$), so $\dot{\kappa} \approx \partial(Kv_s)/\partial s$ and Equation S13 contains the third along-slab derivative of Kv_s. Representing the along-slab variation of Kv_s by a single length L_K, and taking $v_s \approx v_c$,

$$\frac{dQ_s}{ds} \approx \frac{\eta H^3 K v_c}{3L_K^3}, \qquad (S14)$$

which is Equation 12 of the main text. Dimensionally, η [Pa s] $\times$ K [m^{-1}] $\times$ v_c [m s^{-1}] $\times$ H^3/L_K^3 is a stress, as dQ_s/ds must be. This is the bending-dominated stress balance of Ribe (2001, Section 3), in which an applied normal or tangential stress is carried by $\eta H^3\omega/L^3$. L_K is fitted to the models (Figure S-LK; 200 km); it is about twice the local curvature length $K/|dK/ds|$ of the modelled slabs (60–140 km), and much shorter than the length over which v_s varies ($\sim$ 4500 km), so $\dot{\kappa} \approx v_s\,dK/ds$. In the models ω tracks $-Kv_s$ with a slope of 0.75 overall and 0.86–0.89 for the 100 and 130 km slabs, the difference being the slabs' normal motion. The factor 3 is inherited from Equation S12 and is kept so that L_K retains its meaning as the length over which Kv_s varies; it is not an independent fitted constant. Fitting the exponent of K across the suite gives 0.65 (model bootstrap 0.4–0.7), so a single L_K independent of curvature is the right description; the alternative $L_K \propto 1/K$, which would make the dependence K^4, is excluded.

The ratio of the two endmembers, $(\eta H^3 K v_c/3L_K^3)/(\eta H K v_c/L) = H^2 L/(3L_K^3)$, shows why they cannot be told apart at a single plate thickness: at $H = 100$ km, $L_K = 200$ km and L of order a few thousand kilometres, the ratio is of order one, and either form can be fitted to the $H = 100$ km models with its own length. Only the thickness dependence separates them (Figure S-varH).